

PART IV. PROBLEMS, HINTS, SOLUTIONS

This last part offers the reader additional opportunity for practice.

The problems require no more preliminary knowledge than the reader could have acquired from a good high-school curriculum. Yet they are not too easy and not mere routine problems; some of them demand originality and ingenuity.¹²

The hints offer indications leading to the result, mostly by quoting an appropriate sentence from the list; to a very attentive reader ready to pick up suggestions they may reveal the key idea of the solution.

The solutions bring not only the answer but also the procedure leading to the answer, although, of course, the reader has to supply some of the details. Some solutions try to open up some further outlook by a few words placed at the end.

The reader who has earnestly tried to solve the problem has the best chance to profit by the hint and the solution. If he obtains the result by his own means, he may learn something by comparing his method with the method given in print. If, after a serious effort, he is inclined to give up, the hint may supply him with the

¹² Except Problem 1 (widely known, but too amusing to miss) all the problems are taken from the Stanford University Competitive Examinations in Mathematics (there are a few minor changes). Some of the problems were formerly published in *The American Mathematical Monthly* and/or *The California Mathematics Council Bulletin*. In the latter periodical also some solutions were published by the author; they appear appropriately rearranged in the sequel.

missing idea. If even the hint does not help, he may look at the solution, try to isolate the key idea, put the book aside, and then try to work out the solution.

PROBLEMS

1. A bear, starting from the point P , walked one mile due south. Then he changed direction and walked one mile due east. Then he turned again to the left and walked one mile due north, and arrived exactly at the point P he started from. What was the color of the bear?

2. Bob wants a piece of land, exactly level, which has four boundary lines. Two boundary lines run exactly north-south, the two others exactly east-west, and each boundary line measures exactly 100 feet. Can Bob buy such a piece of land in the U.S.?

3. Bob has 10 pockets and 44 silver dollars. He wants to put his dollars into his pockets so distributed that each pocket contains a different number of dollars. Can he do so?

4. To number the pages of a bulky volume, the printer used 2989 digits. How many pages has the volume?

5. Among Grandfather's papers a bill was found:

72 turkeys \$ _67.9_

The first and last digit of the number that obviously represented the total price of those fowls are replaced here by blanks, for they have faded and are now illegible.

What are the two faded digits and what was the price of one turkey?

6. Given a regular hexagon and a point in its plane. Draw a straight line through the given point that divides the given hexagon into two parts of equal area.

7. Given a square. Find the locus of the points from

which the square is seen under an angle (a) of 90° (b) of 45° . (Let P be a point outside the square, but in the same plane. The smallest angle with vertex P containing the square is the "angle under which the square is seen" from P .) Sketch clearly both loci and give a full description.

8. Call "axis" of a solid a straight line joining two points of the surface of the solid and such that the solid, rotated about this line through an angle which is greater than 0° and less than 360° coincides with itself.

Find the axes of a cube. Describe clearly the location of the axes, find the angle of rotation associated with each. Assuming that the edge of the cube is of unit length, compute the arithmetic mean of the lengths of the axes.

9. In a tetrahedron (which is not necessarily regular) two opposite edges have the same length a and they are perpendicular to each other. Moreover they are each perpendicular to a line of length b which joins their midpoints. Express the volume of the tetrahedron in terms of a and b , and prove your answer.

10. The vertex of a pyramid opposite the base is called the *apex*. (a) Let us call a pyramid "isosceles" if its apex is at the same distance from all *vertices* of the base. Adopting this definition, prove that the base of an isosceles pyramid is *inscribed* in a circle the center of which is the foot of the pyramid's altitude.

(b) Now let us call a pyramid "isosceles" if its apex is at the same (perpendicular) distance from all sides of the base. Adopting this definition (different from the foregoing) prove that the base of an isosceles pyramid is *circumscribed* about a circle the center of which is the foot of the pyramid's altitude.

11. Find x , y , u , and v , satisfying the system of four equations

$$\begin{aligned}
 x + 7y + 3v + 5u &= 16 \\
 8x + 4y + 6v + 2u &= -16 \\
 2x + 6y + 4v + 8u &= 16 \\
 5x + 3y + 7v + u &= -16
 \end{aligned}$$

(This may look long and boring: look for a short cut.)

12. Bob, Peter, and Paul travel together. Peter and Paul are good hikers; each walk p miles per hour. Bob has a bad foot and drives a small car in which two people can ride, but not three; the car covers c miles per hour. The three friends adopted the following scheme: They start together, Paul rides in the car with Bob, Peter walks. After a while, Bob drops Paul, who walks on; Bob returns to pick up Peter, and then Bob and Peter ride in the car till they overtake Paul. At this point they change: Paul rides and Peter walks just as they started and the whole procedure is repeated as often as necessary.

(a) How much progress (how many miles) does the company make per hour?

(b) Through which fraction of the travel time does the car carry just one man?

(c) Check the extreme cases $p = 0$ and $p = c$.

13. Three numbers are in arithmetic progression, three other numbers in geometric progression. Adding the corresponding terms of these two progressions successively, we obtain

$$85, \quad 76, \quad \text{and} \quad 84$$

respectively, and, adding all three terms of the arithmetic progression, we obtain 126. Find the terms of both progressions.

14. Determine m so that the equation in x

$$x^4 - (3m + 2)x^2 + m^2 = 0$$

has four real roots in arithmetic progression.

15. The length of the perimeter of a right triangle is 60 inches and the length of the altitude perpendicular to the hypotenuse is 12 inches. Find the sides.

16. From the peak of a mountain you see two points, A and B , in the plain. The lines of vision, directed to these points, include the angle γ . The inclination of the first line of vision to a horizontal plane is α , that of the second line β . It is known that the points A and B are on the same level and that the distance between them is c .

Express the elevation x of the peak above the common level of A and B in terms of the angles α , β , γ , and the distance c .

17. Observe that the value of

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!}$$

is $1/2$, $5/6$, $23/24$ for $n = 1, 2, 3$, respectively, guess the general law (by observing more values if necessary) and prove your guess.

18. Consider the table

1	=	1
3 + 5	=	8
7 + 9 + 11	=	27
13 + 15 + 17 + 19	=	64
21 + 23 + 25 + 27 + 29	=	125

Guess the general law suggested by these examples, express it in suitable mathematical notation, and prove it.

19. The side of a regular hexagon is of length n (n is an integer). By equidistant parallels to its sides the hexagon is divided into T equilateral triangles each of which has sides of length 1. Let V denote the number of vertices appearing in this division, and L the number of boundary lines of length 1. (A boundary line belongs to one or two triangles, a vertex to two or more triangles.) When

$n = 1$, which is the simplest case, $T = 6$, $V = 7$, $L = 12$. Consider the general case and express T , V , and L in terms of n . (Guessing is good, proving is better.)

20. In how many ways can you change one dollar? (The "way of changing" is determined if it is known how many coins of each kind—cents, nickels, dimes, quarters, half dollars—are used.)

HINTS

1. *What is the unknown?* The color of a bear—but how could we find the color of a bear from mathematical data? *What is given?* A geometrical situation—but it seems self-contradictory: how could the bear, after walking three miles in the manner described, return to his starting point?

2. *Do you know a related problem?*

3. If Bob had very many dollars, he would have obviously no difficulty in filling each of his pockets differently. *Could you restate the problem?* What is the minimum number of dollars that can be put in 10 pockets so that no two different pockets contain the same amount?

4. *Here is a problem related to yours:* If the book has exactly 9 numbered pages, how many digits uses the printer? (9, of course.) Here is another *problem related to yours:* If the book has exactly 99 numbered pages, how many digits does the printer use?

5. *Could you restate the problem?* What can the two faded digits be if the total price, expressed in cents, is divisible by 72?

6. *Could you imagine a more accessible related problem? A more general problem? An analogous problem?* (GENERALIZATION, 2.)

7. *Do you know a related problem?* The locus of the points from which a given segment of a straight line is

seen under a given angle consists of two circular arcs, ending in the extreme points of the segment, and symmetric to each other with respect to the segment.

8. I assume that the reader is familiar with the shape of the cube and has found certain axes just by inspection—but are they *all* the axes? *Can you prove* that your list of axes is exhaustive? Has your list a clear principle of classification?

9. *Look at the unknown!* The unknown is the volume of a tetrahedron—yes, I know, the volume of any pyramid can be computed when the base and the height are given (product of both, divided by 3) but in the present case neither the base nor the height is given. *Could you imagine a more accessible related problem?* (Don't you see a more accessible tetrahedron which is an aliquot part of the given one?)

10. *Do you know a related theorem?* Do you know a related . . . simpler . . . *analogous theorem?* Yes: the foot of the altitude is the mid-point of the base in an isosceles triangle. Here is a *theorem related to yours and proved before*. *Could you use . . . its method?* The theorem on the isosceles triangle is proved from congruent right triangles of which the altitude is a common side.

11. It is assumed that the reader is somewhat familiar with systems of linear equations. To solve such a system, we have to combine its equations in some way—look out for relations between the equations which could indicate a particularly advantageous combination.

12. *Separate the various parts of the condition. Can you write them down?* Between the start and the point where the three friends meet again there are three different phases:

- (1) Bob rides with Paul
- (2) Bob rides alone
- (3) Bob rides with Peter.

Call t_1 , t_2 , and t_3 the durations of these phases, respectively. How could you split the condition into appropriate parts?

13. *Separate the various parts of the condition. Can you write them down?* Let

$$a - d, \quad a, \quad a + d$$

be the terms of the arithmetic progression, and

$$bg^{-1}, \quad b, \quad bg$$

be the terms of the geometric progression.

14. *What is the condition?* The four roots must form an arithmetic progression. Yet the equation has a particular feature: it contains only even powers of the unknown x . Therefore, if a is a root, $-a$ is also a root.

15. *Separate the various parts of the condition. Can you write them down?* We may distinguish three parts in the condition, concerning

- (1) perimeter
- (2) right triangle
- (3) height to hypotenuse.

16. *Separate the various parts of the condition. Can you write them down?* Let a and b stand for the lengths of the (unknown) lines of vision, α and β for their inclinations to the horizontal plane, respectively. We may distinguish three parts in the condition, concerning

- (1) the inclination of a
- (2) the inclination of b
- (3) the triangle with sides a , b , and c .

17. Do you recognize the denominators 2, 6, 24? Do you know a related problem? An analogous problem? (INDUCTION AND MATHEMATICAL INDUCTION.)

18. Discovery by induction needs observation. Observe the right-hand sides, the initial terms of the left-hand sides, and the final terms. What is the general law?

19. *Draw a figure.* Its observation may help you to discover the law inductively, or it may lead you to relations between T , V , L , and n .

20. *What is the unknown?* What are we supposed to seek? Even the aim of the problem may need some clarification. *Could you imagine a more accessible related problem? A more general problem? An analogous problem?* Here is a *very* simple analogous problem: In how many ways can you pay one cent? (There is just one way.) Here is a more general problem: In how many ways can you pay the amount of n cents using these five kinds of coins: cents, nickels, dimes, quarters, and half dollars. We are especially concerned with the particular case $n = 100$.

In the simplest particular cases, for small n , we can figure out the answer without any high-brow method, just by trying, by inspection. Here is a short table (which the reader should check).

n	4	5	9	10	14	15	19	20	24	25
E_n	1	2	2	4	4	6	6	9	9	13

The first line lists the amounts to be paid, generally called n . The second line lists the corresponding numbers of "ways of paying," generally called E_n . (Why I have chosen this notation is a secret of mine which I am not willing to give away at this stage.)

We are especially concerned with E_{100} , but there is little hope that we can compute E_{100} without some clear method. In fact the present problem requires a little more from the reader than the foregoing ones; he should *create* a little *theory*.

Our question is general (to compute E_n for general n),

but it is "isolated." *Could you imagine a more accessible related problem? An analogous problem?* Here is a very simple analogous problem: Find A_n , the number of ways to pay the amount of n cents, using only cents. ($A_n = 1$.)

SOLUTIONS

1. You think that the bear was white and the point P is the North Pole? *Can you prove that this is correct?* As it was more or less understood, we idealize the question. We regard the globe as exactly spherical and the bear as a moving material point. This point, moving due south or due north, describes an arc of a *meridian* and it describes an arc of a *parallel circle* (parallel to the equator) when it moves due east. We have to distinguish two cases.

(1) If the bear returns to the point P along a meridian *different* from the one along which he left P , P is necessarily the North Pole. In fact the only other point of the globe in which two meridians meet is the South Pole, but the bear could leave this pole only in moving northward.

(2) The bear could return to the point P along the same meridian he left P if, when walking one mile due east, he describes a parallel circle exactly n times, where n may be 1, 2, 3 . . . In this case P is not the North Pole, but a point on a parallel circle very close to the South Pole (the perimeter of which, expressed in miles, is slightly inferior to $2\pi + 1/n$).

2. We represent the globe as in the solution of Problem 1. The land that Bob wants is bounded by two meridians and two parallel circles. Imagine two fixed meridians, and a parallel circle moving *away* from the equator: the arc on the moving parallel intercepted by the two fixed meridians is steadily shortened. The center of the land that Bob wants should be on the equator: he can *not* get it in the U.S.

3. The least possible number of dollars in a pocket is obviously 0. The next greater number is at least 1, the next greater at least 2 . . . and the number in the last (tenth) pocket is at least 9. Therefore, the number of dollars required is at least

$$0 + 1 + 2 + 3 + \dots + 9 = 45$$

Bob cannot make it: he has only 44 dollars.

4. A volume of 999 numbered pages needs

$$9 + 2 \times 90 + 3 \times 900 = 2889$$

digits. If the bulky volume in question has x pages

$$\begin{aligned} 2889 + 4(x - 999) &= 2989 \\ x &= 1024 \end{aligned}$$

This problem may teach us that a preliminary estimate of the unknown may be useful (or even necessary, as in the present case).

5. If $_679_$ is divisible by 72, it is divisible both by 8 and by 9. If it is divisible by 8, the number $79_$ must be divisible by 8 (since 1000 is divisible by 8) and so $79_$ must be 792: the last faded digit is 2. If $_6792$ is divisible by 9, the sum of its digits must be divisible by 9 (the rule about "casting out nines") and so the first faded digit must be 3. The price of one turkey was (in grandfather's time) $\$367.92 \div 72 = \5.11 .

6. "*A point and a figure with a center of symmetry (in the same plane) are given in position. Find a straight line that passes through the given point and bisects the area of the given figure.*" The required line passes, of course, through the center of symmetry. See INVENTOR'S PARADOX.

7. In any position the two sides of the angle must pass through two vertices of the square. As long as they pass through the same pair of vertices, the angle's vertex

moves along the same arc of circle (by the theorem underlying the hint). Hence each of the two loci required consists of several arcs of circle: of 4 semicircles in the case (a) and of 8 quarter circles in the case (b); see Fig. 31.

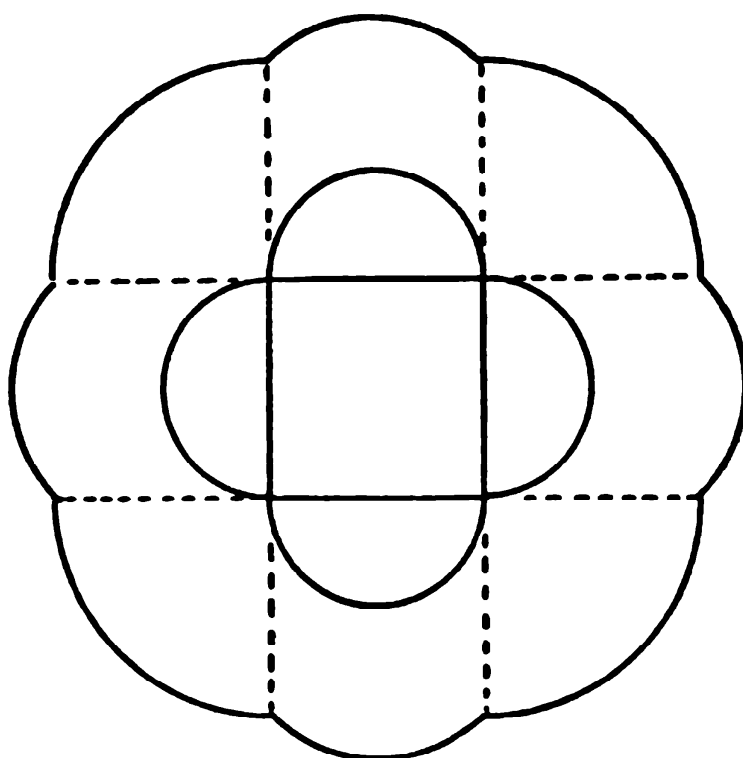


FIG. 31

8. The axis pierces the surface of the cube in some point which is either a vertex of the cube or lies on an edge or in the interior of a face. If the axis passes through a point of an edge (but not through one of its endpoints) this point must be the midpoint: otherwise the edge could not coincide with itself after the rotation. Similarly, an axis piercing the interior of a face must pass through its center. Any axis must, of course, pass through the center of the cube. And so there are three kinds of axes:

- (1) 4 axes, each through two opposite vertices; angles $120^\circ, 240^\circ$

- (2) 6 axes, each through the mid-points of two opposite edges; angle 180°
 (3) 3 axes, each through the center of two opposite faces; angles 90° , 180° , 270° .

For the length of an axis of the first kind see section 12; the others are still easier to compute. The desired average is

$$\frac{4\sqrt{3} + 6\sqrt{2} + 3}{13} = 1.416.$$

(This problem may be useful in preparing the reader for the study of crystallography. For the reader sufficiently advanced in the integral calculus it may be observed that the average computed is a fairly good approximation to the "average width" of the cube, which is, in fact, $3/2 = 1.5$.)

9. The plane passing through one edge of length a and the perpendicular of length b divides the tetrahedron into two *more accessible* congruent tetrahedra, each with base $ab/2$ and height $a/2$. Hence the required volume

$$= 2 \cdot \frac{1}{3} \cdot \frac{ab}{2} \cdot \frac{a}{2} = \frac{a^2b}{6}.$$

10. The base of the pyramid is a polygon with n sides. In the case (a) the n lateral edges of the pyramid are equal; in the case (b) the altitudes (drawn from the apex) of its n lateral faces are equal. If we draw the altitude of the pyramid and join its foot to the n vertices of the base in the case (a), but to the feet of the altitudes of the n lateral faces in the case (b), we obtain, in both cases, n *right triangles of which the altitude* (of the pyramid) *is a common side*: I say that these n right triangles are congruent. In fact the hypotenuse [a lateral edge in the case (a), a lateral altitude in the case (b)] is of the same length in each, according to the definitions

laid down in the proposed problem; we have just mentioned that another side (the altitude of the pyramid) and an angle (the right angle) are common to all. In the n congruent triangles the third sides must also be equal; they are drawn from the same point (the foot of the altitude) in the same plane (the base): they form n radii of a circle which is circumscribed about, or inscribed into, the base of the pyramid, in the cases (a) and (b), respectively. [In the case (b) it remains to show, however, that the n radii mentioned are perpendicular to the respective sides of the base; this follows from a well-known theorem of solid geometry on projections.]

It is most remarkable that a plane figure, the isosceles triangle, may have *two different analogues* in solid geometry.

11. Observe that the first equation is so related to the last as the second is to the third: the coefficients on the left-hand sides are the same, but in opposite order, whereas the right-hand sides are opposite. Add the first equation to the last and the second to the third:

$$\begin{aligned} 6(x + u) + 10(y + v) &= 0, \\ 10(x + u) + 10(y + v) &= 0. \end{aligned}$$

This can be regarded as a system of two linear equations for two unknowns, namely for $x + u$ and $y + v$, and easily yields

$$x + u = 0, \quad y + v = 0.$$

Substituting $-x$ for u and $-y$ for v in the first two equations of the original system, we find

$$\begin{aligned} -4x + 4y &= 16 \\ 6x - 2y &= -16. \end{aligned}$$

This is a simple system which yields

$$x = -2, \quad y = 2, \quad u = 2, \quad v = -2$$

12. Between the start and the meeting point each of the friends traveled the same distance. (Remember, distance = velocity $\times$ time.) We distinguish two parts in the condition:

Bob traveled as much as Paul:

$$ct_1 - ct_2 + ct_3 = ct_1 + pt_2 + pt_3,$$

Paul traveled as much as Peter:

$$ct_1 + pt_2 + pt_3 = pt_1 + pt_2 + ct_3.$$

The second equation yields

$$(c - p)t_1 = (c - p)t_3.$$

We assume, of course, that the car travels faster than a pedestrian, $c > p$. It follows

$$t_1 = t_3;$$

that is, Peter walks just as much as Paul. From the first equation, we find that

$$\frac{t_3}{t_2} = \frac{c + p}{c - p}$$

which is, of course, also the value for t_1/t_2 . Hence we obtain the answers:

$$(a) \quad \frac{c(t_1 - t_2 + t_3)}{t_1 + t_2 + t_3} = \frac{c(c + 3p)}{3c + p}$$

$$(b) \quad \frac{t_2}{t_1 + t_2 + t_3} = \frac{c - p}{3c + p}$$

(c) In fact, $0 < p < c$. There are two extreme cases:

If $p = 0$ (a) yields $c/3$ and (b) yields $1/3$

If $p = c$ (a) yields c and (b) yields 0 .

These results are easy to see without computation.

13. The condition is easily split into four parts expressed by the four equations

$$\begin{aligned} a - d + bg^{-1} &= 85 \\ a + b &= 76 \\ a + d + bg &= 84 \\ 3a &= 126. \end{aligned}$$

The last equation yields $a = 42$, then the second $b = 34$. Adding the remaining two equations (to eliminate d), we obtain

$$2a + b(g^{-1} + g) = 169.$$

Since a and b are already known, we have here a quadratic equation for g . It yields

$$g = 2, \quad d = -26 \quad \text{or} \quad g = 1/2, \quad d = 25.$$

The progressions are

$$\begin{array}{ccc} 68, 42, 16 & & 17, 42, 67 \\ & \text{or} & \\ 17, 34, 68 & & 68, 34, 17 \end{array}$$

14. If a and $-a$ are the roots having the least absolute value, they will stand next to each other in the progression which will, therefore, be of the form

$$-3a, -a, a, 3a.$$

Hence the left-hand side of the proposed equation must have the form

$$(x^2 - a^2)(x^2 - 9a^2).$$

Carrying out the multiplication and comparing coefficients of like powers, we obtain the system

$$\begin{aligned} 10a^2 &= 3m + 2, \\ 9a^4 &= m^2. \end{aligned}$$

Elimination of a yields

$$19m^2 - 108m - 36 = 0.$$

Hence $m = 6$ or $-6/19$.

15. Let a , b , and c denote the sides, the last being the hypotenuse. The three parts of the condition are expressed by

$$\begin{aligned} a + b + c &= 60 \\ a^2 + b^2 &= c^2 \\ ab &= 12c. \end{aligned}$$

Observing that

$$(a + b)^2 = a^2 + b^2 + 2ab$$

we obtain

$$(60 - c)^2 = c^2 + 24c.$$

Hence $c = 25$ and either $a = 15$, $b = 20$ or $a = 20$, $b = 15$ (no difference for the triangle).

16. The three parts of the condition are expressed by

$$\sin \alpha = \frac{x}{a},$$

$$\sin \beta = \frac{x}{b},$$

$$c^2 = a^2 + b^2 - 2ab \cos \gamma$$

The elimination of a and b yields

$$x^2 = \frac{c^2 \sin^2 \alpha \sin^2 \beta}{\sin^2 \alpha + \sin^2 \beta - 2 \sin \alpha \sin \beta \cos \gamma}.$$

17. We conjecture that

$$\frac{1}{2!} + \frac{2}{3!} + \cdots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}.$$

Following the pattern of INDUCTION AND MATHEMATICAL INDUCTION, we ask: Does the conjectured formula remain

true when we pass from the value n to the next value $n + 1$? Along with the formula above we should have

$$\frac{1}{2!} + \frac{2}{3!} + \cdots + \frac{n}{(n+1)!} + \frac{n+1}{(n+2)!} = 1 - \frac{1}{(n+2)!}$$

Check this by subtracting from it the former:

$$\frac{n+1}{(n+2)!} = -\frac{1}{(n+2)!} + \frac{1}{(n+1)!}$$

which boils down to

$$\frac{n+2}{(n+2)!} = \frac{1}{(n+1)!}$$

and this last equation is obviously true for $n = 1, 2, 3, \dots$ hence, by following the pattern referred to above, we can prove our conjecture.

18. In the n th line the right-hand side seems to be n^3 and the left-hand side a sum of n terms. The final term of this sum is the m th odd number, or $2m - 1$, where

$$m = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2};$$

see INDUCTION AND MATHEMATICAL INDUCTION, 4. Hence the final term of the sum on the left-hand side should be

$$2m - 1 = n^2 + n - 1.$$

We can derive hence the initial term of the sum considered in *two* ways: going back $n - 1$ steps from the final term, we find

$$(n^2 + n - 1) - 2(n - 1) = n^2 - n + 1$$

whereas, advancing one step from the final term of the foregoing line, we find

$$[(n - 1)^2 + (n - 1) - 1] + 2$$

which, after routine simplification, boils down to the same: good! We assert therefore that

$$(n^2 - n + 1) + (n^2 - n + 3) + \cdots + (n^2 + n - 1) = n^3$$

where the left-hand side indicates the sum of n successive terms of an arithmetic progression the difference of which is 2. If the reader knows the rule for the sum of such a progression (arithmetic mean of the initial term and the final term, multiplied by the number of terms), he can verify that

$$\frac{(n^2 - n + 1) + (n^2 + n - 1)}{2} n = n^3$$

and so prove the assertion.

(The rule quoted can be easily proved by a picture little different from Fig. 18.)

19. The length of the perimeter of the regular hexagon with side n is $6n$. Therefore, this perimeter consists of $6n$ boundary lines of length 1 and contains $6n$ vertices. Therefore, in the transition from $n - 1$ to n , V increases by $6n$ units, and so

$$V = 1 + 6(1 + 2 + 3 + \cdots + n) = 3n^2 + 3n + 1;$$

see INDUCTION AND MATHEMATICAL INDUCTION, 4. By 3 diagonals through its center the hexagon is divided into 6 (large) equilateral triangles. By inspection of one of these

$$T = 6(1 + 3 + 5 + \cdots + 2n - 1) = 6n^2$$

(rule for the sum of an arithmetic progression, quoted in the solution of Problem 18). The T triangles have jointly $3T$ sides. In this total $3T$ each internal line of division of length 1 is counted twice, whereas the $6n$ lines along the perimeter of the hexagon are counted but once. Hence

$$2L = 3T + 6n, \quad L = 9n^2 + 3n.$$

(For the more advanced reader: it follows from Euler's theorem on polyhedra that $T + V = L + 1$. Verify this relation!)

20. Here is a well-ordered array of analogous problems: Compute A_n , B_n , C_n , D_n and E_n . Each of these quantities represents the number of ways to pay the amount of n cents; the difference is in the coins used:

A_n only cents

B_n cents and nickels

C_n cents, nickels, and dimes

D_n cents, nickels, dimes, and quarters

E_n cents, nickels, dimes, quarters, and half dollars.

The symbols E_n (reason now clear) and A_n were used before.

All ways and manners to pay the amount of n cents with the five kinds of coin are enumerated by E_n . We may, however, distinguish two possibilities:

First. No half dollar is used. The number of such ways to pay is D_n , by definition.

Second. A half dollar (possibly more) is used. After the first half dollar is laid on the counter, there remains the amount of $n - 50$ cents to pay, which can be done in exactly E_{n-50} ways.

We infer that

$$E_n = D_n + E_{n-50}.$$

Similarly

$$D_n = C_n + D_{n-25},$$

$$C_n = B_n + C_{n-10},$$

$$B_n = A_n + B_{n-5}.$$

A little attention shows that these formulas remain valid if we set

$$A_0 = B_0 = C_0 = D_0 = E_0 = 1$$

(which obviously makes sense) and regard any one of the quantities $A_n, B_n \dots E_n$ as equal to 0 when its subscript happens to be negative. (For example, $E_{25} = D_{25}$, as can be seen immediately, and this agrees with our first formula since $E_{25-50} = E_{-25} = 0$.)

Our formulas allow us to compute the quantities considered *recursively*, that is, by going back to lower values of n or to former letters of the alphabet. For example, we can compute C_{30} by simple addition if C_{20} and B_{30} are already known. In the table below the initial row, headed by A_n , and the initial column, headed by 0, contain only numbers equal to 1. (Why?) Starting from these initial numbers, we compute the others recursively, by simple additions: any other number of the table is equal either to the number above it or to the sum of two numbers: the number above it and another at the proper distance to the left. For example,

$$C_{30} = B_{30} + C_{20} = 7 + 9 = 16$$

The computation is carried through till $E_{50} = 50$: *you can pay 50 cents in exactly 50 different ways*. Carrying it further, the reader can convince himself that $E_{100} = 292$: *you can change a dollar in 292 different ways*.

n	0	5	10	15	20	25	30	35	40	45	50
A_n	1	1	1	1	1	1	1	1	1	1	1
B_n	1	2	3	4	5	6	7	8	9	10	11
C_n	1	2	4	6	9	12	16	20	25	30	36
D_n	1	2	4	6	9	13	18	24	31	39	49
E_n	1	2	4	6	9	13	18	24	31	39	50